

Selective Invariance Violations in Large Language Model Moral Judgment: Validated Moral Geometry, Verdict-Level Attacks, and an Invariance Defense

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Abstract—Large language models increasingly gate safety-critical decisions, yet an attacker who controls only the *presentation* of facts—not the facts themselves—can displace their moral judgments. Our conference paper demonstrated this attack: across five models, salience manipulations (linguistic framing, emotional anchoring, irrelevant sensory detail) reliably displace judgments in a 7-dimensional harm space, euphemistic rewriting silently reclassifies flagged content past a moderation threshold, and the displacement propagates through a rule-based decision kernel, flipping typed verdicts. This extended article completes the arc from attack to defense. First, we *validate the moral geometry* beyond inter-model reliability ($\text{ICC}(2,k)=0.97$): a pre-registered discovery-and-admission loop flags three candidate missing dimensions, validates one (`identity_attack`; cross-dataset held-out AUROC 0.80, CI [0.78, 0.83]), retracts one, and declines one—demonstrating that the dimension set is extensible and falsifiable, not asserted. A pre-registered bifactor readout sharpens what “validated” means: the coordinate system is one strong general-valence factor plus a few separable specifics—five nominal axes are ≥ 0.98 predictable from general valence, while privacy, autonomy, and environmental are genuinely their own—so the framework distinguishes validated *transfer* from validated *specificity*. Second, we characterize *which* axes are robust measurement targets: valence encoders are register-invariant under domain-adversarial training while foundation-presence encoders are register-bound, a dissociation confirmed by three independent methods. Third, we present a *measured defense*: equivalence-class averaging over a generated paraphrase class halves decision-layer drift ($0.407 \rightarrow 0.219$, meeting a pre-registered bar of 0.5) and a non-refusing back-translation red-team shows the residual hole belongs to the paraphrase generator, not the mechanism. Fourth, a leakage-controlled learned contraction (out-of-fold AUROC 0.863) lets the pipeline *moderate*, not merely escalate, and the invariance property extends cross-lingually (index 0.72–0.80 across five languages, harmful $\approx$ benign). The result is an end-to-end, audit-proof content-moderation instrument in which every scored dimension carries validation provenance, and every decision is invariant under re-description or explicitly escalated.

Index Terms—LLM security, adversarial evaluation, moral judgment, invariance testing, vulnerability profiling, AI safety, content moderation, sentence embeddings, audit trails

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I. INTRODUCTION

When a language model judges a moral scenario, its output can be understood as a point in a multi-dimensional harm space. A secure evaluator should produce the same point regardless of how the scenario is *described*: euphemistic or dramatic register, irrelevant sensory detail, social pressure. Failure of this property is a *behavioral manipulation vulnerability*: an attacker shifts the decision by modifying morally irrelevant surface features without changing the underlying facts.

Our conference paper [1] established the attack. Across five models and two architecture families, three perturbation types that manipulate perceptual salience reliably displace judgments in a 7-dimensional harm space (Cohen’s $d = 0.60$ – 1.06 for emotional anchoring), while two that do not manipulate salience (gender swap, evaluation order) produce no displacement. A worked exploit showed the deployment consequence: euphemistic rewriting lowers panel-averaged harm by 14.0 points on a 0–70 scale, silently moving three of six audited gold items from FLAG to PASS at a median-calibrated threshold; routed through the rule-based DEME v3 decision kernel [2], the same rewriting flips a restrictive typed verdict to a permissive one in 13.7% of 161 scenario–model cases. Both reviewers’ principal concern was validation: is the harm space itself reliable, or asserted?

This extended article answers that question and completes the arc from attack to defense. It makes four contributions beyond [1]:

- 1) **A validated, extensible, falsifiable moral geometry (Sec. IV).** Beyond the inter-model reliability panel ($\text{ICC}(2,k)=0.969$), we operate a pre-registered *discovery-and-admission loop* on the learned-encoder instantiation of the same geometry: residual analysis flags candidate missing dimensions, and a shared admission gate with pre-registered nulls decides. Of three flagged candidates, one is validated (`identity_attack`: cross-dataset held-out AUROC 0.80, CI [0.78, 0.83], $+0.25$ over its pre-registered null), one is retracted (`threat`), and one is declined (`sexual_content`, shown to be a policy-norms signal rather than a moral axis). A framework that rejects its own candidates cannot be accused of asserting its coordinates. A pre-registered bifactor readout then separates general moral valence

from axis-specific signal: the geometry is one strong shared factor plus a handful of separable specifics, so axes that clear the gate on *transfer* are distinguished from those with validated *specificity*. We further characterize which axes are robust measurement targets: *valence* encoders are register-invariant, while foundation-*presence* encoders are register-bound—a dissociation confirmed by three independent methods.

- 2) **A measured invariance defense (Sec. VI).** Equivalence-class averaging—generate the input’s paraphrase class, average per-dimension perception over the class, then decide—halves decision-layer drift at scale (raw 0.407 $\rightarrow$ 0.219 on a pre-registered split-half protocol, meeting the pre-registered bar $\theta_d \leq 0.5$). A red-team leg with a non-refusing back-translation paraphraser closes the LLM generator’s 24% refusal hole ($\theta_d = 0.301$ on harmful content), locating the residual weakness in the generator, not the mechanism. The defense’s trust boundary is made explicit and auditable.
- 3) **From escalation to moderation (Sec. VII).** A leakage-controlled learned contraction over the validated perception channels (out-of-fold AUROC 0.863, +0.084 over the pre-extension baseline, with `identity_attack` the dominant feature) lets the pipeline make allow/remove decisions where validated and escalate where not.
- 4) **Cross-lingual generalization (Sec. VIII).** The invariance property is located architecturally in the canonicalization layer and confirmed at scale across five languages and four scripts (invariance index 0.72–0.80, harmful $\approx$ benign), converting a stated limitation of [1] into a measured result.

[TODO: Contribution 5, once run: native 9-D/10-D re-run of the five vulnerability tracks through the erisml-compiler pipeline, measuring displacement in the full DEME space and at the verdict level per track (outline Part C; NRP open panel; planned August 2026).]

A. Two Instantiations, One Geometry

A methodological point governs the whole article, and we state it before any result. The conference paper measures displacement of *LLM-as-judge* scores: a model under test emits the 7-D harm vector directly. The new results in Secs. IV–VIII were obtained on a *learned-encoder perception pipeline*: small per-dimension encoders (the `xbse` library [3]), each validated through a shared, pre-registered cross-dataset gate, feeding the same DEME decision kernel through the Moral Spectrum Analyzer orchestration layer [4]. These are two instantiations of one geometric framework, not one experiment: the former audits deployed judges; the latter is the hardened instrument the audit motivates. The unification is the DEME v3 moral vector both emit, and the native re-run of the five vulnerability tracks through the learned-encoder pipeline is the designed experiment that closes the loop (**[TODO: Sec. IX]**). Conflating the two instantiations would re-open the validation critique this article answers; we keep the distinction explicit throughout.

II. RELATED WORK

Sycophancy, framing, and cognitive biases in LLMs. Sharma et al. [5] and Perez et al. [6] established sycophantic response shifts; Tversky and Kahneman [7] established framing effects in humans and Echterhoff et al. [8] surveyed their LLM analogues; Scherrer et al. [9] measured prompt sensitivity of encoded moral beliefs. These probe one perturbation at a time and emit scalar robustness scores; our framework applies qualitatively distinct perturbations in a shared coordinate system and profiles vulnerability per axis [1].

Randomized smoothing and perturbation-averaged defenses. Our equivalence-class averaging defense (Sec. VI) is a semantic analogue of randomized smoothing [12], which certifies a classifier by aggregating predictions over random input perturbations, and of SmoothLLM [13], which averages over character-level perturbations to defeat jailbreaks. The differences are the perturbation class (meaning-preserving paraphrase, generated by a model that thereby joins the trust boundary) and the certified quantity (invariance of a multi-dimensional moral decision, not a binary jailbreak flag). Character- or token-level smoothing cannot express “same facts, different salience,” which is precisely the attack surface identified in [1].

Toxicity detection and its blind spots. Perspective-style toxicity scoring [14] and Detoxify-class models compress a morally significant decision into one scalar; documented identity-term biases motivated the `identity_attack` attribute in the civil-comments corpus [14] and the Measuring Hate Speech corpus [15]. Our contribution is not the label but the *instrument*: a discovery loop that finds an unscored dimension in a fixed taxonomy, a pre-registered gate that validates it cross-dataset, and an audit trail that carries its validation provenance into every decision (Sec. IV-B).

Moral Foundations Theory and moral corpora. MFT’s binding foundations (loyalty, sanctity/purity) [16] are exactly the axes a harm-centric taxonomy under-covers; we test whether they are learnable, transferable signals using Social-Chemistry-101 [17], ETHICS [18], and the Moral Foundations Reddit Corpus [19] (Sec. IV-C).

Multilingual representation. LaBSE [20] and BGE-M3 [21] provide the multilingual embedding spaces our canonicalization layer builds on; NLLB [22] provides the translation and back-translation machinery for the cross-lingual and red-team experiments. Thiele [11] showed the moral dimensions of the geometric-ethics framework [10] are linearly decodable from LaBSE embeddings; the `xbse` encoders operationalize that finding as validated per-dimension instruments [3].

III. THE MEASUREMENT INSTRUMENT

A. Harm Space, DEME v3, and the 10th Channel

The conference framework scores moral judgment in a 7-D harm space $\mathcal{H} = (h_{\text{phys}}, h_{\text{emot}}, h_{\text{fin}}, h_{\text{auto}}, h_{\text{trust}}, h_{\text{soc}}, h_{\text{ident}})$, a measurement-reliable projection of the 9-D DEME v3 `MoralVector` of the ErisML compiler [2] (correspondence table in [1]). The learned-encoder instantiation scores the

DEME axes directly: one small encoder per dimension, fine-tuned from BGE-M3 [21] by a shared contrastive objective, each emitting a signed per-axis score [3].

The discovery result of Sec. IV-B adds a tenth channel. Because the compiler’s `MoralTensorV3` is a frozen 9-axis type (validating `shape[0]==9`), the validated `identity_attack` dimension rides as a first-class *extension channel* in tensor metadata—not a compiler fork—and the moral vector, decision inputs, and hash-chained audit proofs all span 10 axes. This supersedes the conference paper’s weakest correspondence-table cell, where h_{idnt} mapped only partially onto privacy/standing: identity-directed attack is now a validated, separately-scored axis.

B. Pipeline and Audit Trail

The deployed pipeline (the Moral Spectrum Analyzer [4]) composes: perception (validated `xbse` encoders, or a replay cache of their outputs; an unvalidated stub backend is labeled as such and never presented as a real score) → canonicalization (Sec. VIII) → DEME tensor → decision (learned contraction where validated, escalation where not; Sec. VII) → hash-chained, re-verifiable audit proof. Every dimension score carries validation provenance; every decision records the evidence—including, for the invariance mechanism of Sec. VI, the paraphrase-class member hashes—needed to re-verify it.

IV. VALIDATING THE MORAL GEOMETRY

Both reviewers of [1] pressed one point: the harm space is “asserted more than validated.” The camera-ready answered with reliability: across an open panel of six models from five families, the seven dimensions are measured with $\text{ICC}(2,k)=0.969$ and Krippendorff $\alpha = 0.836$ (test–retest median $r = 0.96$) [1]. Reliability, however, only shows the coordinates are *consistently readable*. This section shows they are *extensible* (missing dimensions can be discovered and admitted), *falsifiable* (candidates can fail admission and validated axes can be retired), and *characterized* (we know which axes are robust measurement targets and which are register-bound).

A. The Validation Gate

Every learned encoder must clear one shared, pre-registered gate before its scores may be used downstream: held-out *cross-dataset* transfer (structure AUROC on a second, independent corpus), margin over a pre-registered null (untrained baseline and bag-of-words control), and a fuzz test (surface perturbations must move the score much less than moral content). Bars are derived from the data and registered before results are seen; an encoder that fails is reported as failed and its axis is escalated, not scored. On the nine original DEME axes, eight of nine encoders cleared the gate; `rights_respect` failed and is carried as unvalidated—the framework reports its own gap [3].

B. Discovery: the Admission Loop Has Teeth

A fixed taxonomy cannot report what it fails to see. The discovery loop mines the residual structure of moderation corpora for signal the current axes do not carry, flags candidate dimensions, and submits each to the admission gate of Sec. IV-A. Three candidates were flagged, each with bootstrap confidence intervals on its residual lift:

TABLE I
DISCOVERY-AND-ADMISSION SCORECARD. RESIDUAL LIFT IS THE BOOTSTRAP-CI’D IMPROVEMENT IN EXPLAINED MODERATION SIGNAL BEYOND THE EXISTING AXES AT FLAG TIME.

Candidate	Lift	Gate outcome	Disposition
<code>identity_attack</code>	+0.24	AUROC 0.80 [0.78, 0.83]	validated
<code>threat</code>	—	failed balanced resample	retracted
<code>sexual_content</code>	+0.20	policy signal, not moral	declined

Validated: `identity_attack`. Trained jointly on two independent corpora (Jigsaw civil-comments `identity_attack` labels [14] and the Berkeley Measuring Hate Speech continuous score [15]; $n = 6,400$), the encoder reaches cross-dataset held-out AUROC 0.80 (anchor-bootstrap CI [0.78, 0.83]), +0.25 over its pre-registered null, with fuzz ratio 14.3. It is wired live as the 10th channel (Sec. III). Behavior is specific: hate incitement scores strongly negative (−0.50) while journalistic reporting of hate remains near-neutral (+0.18); euphemized attacks evade the raw feeder (+0.36)—which is precisely the decision-layer invariance case Sec. VI addresses. We own the label overlap plainly: `identity_attack` is an existing Perspective-style attribute [14]; the novelty is the *instrument*—discovery from residuals, pre-registered admission, provenance-carrying deployment—not the label.

Retracted: `threat`. The flagged lift did not survive a balanced resample; the candidate was retracted before build.

Declined: `sexual_content`. Assessment showed the flagged signal is a *policy-norms* channel, not a moral axis: in civil-comments, explicit content is 87.7% toxic vs. 1.4% non-toxic—the corpus contains almost no non-harassing explicit content to carry an independent moral valence, so the apparent moral weight is harassment (already covered). In Social-Chemistry [17], sexual moral valence attaches to *conduct and consent* (autonomy, care), not to explicitness itself; consensual explicitness is not a moral violation. The dimension is routed as a platform-configurable policy channel *outside* the moral tensor.

One validated, one retracted, one declined: the admission criterion has teeth, and the geometry’s dimension set is an empirical claim with a falsification record, not an assertion.

C. Binding Foundations: Loyalty and Purity

Moral Foundations Theory predicts that a harm-centric taxonomy under-covers the “binding” foundations [16]. We tested whether loyalty/betrayal and sanctity/degradation are learnable, transferable valence signals under the same gate: training pairs from Social-Chemistry-101 (loyalty-betrayal 41k, sanctity-degradation 15k, signed by action moral judgment) [17] crossed with ETHICS keyword scenarios [18],

with the Moral Foundations Reddit Corpus [19] as a presence-only cross-check. Both pass: loyalty AUROC 0.911 (+0.406 over null); purity 0.811—+0.156 over a deliberately hostile disgust-lexicon bag-of-words null of 0.656, so the encoder learns beyond keywords. MFRC presence cross-checks land at 0.608/0.635. Caveats are pre-registered and disclosed: the ETHICS second corpus is thin (602/337 rows), so the purity AUROC is provisional; and passing the transfer gate does not establish independence from the general moral-valence factor—the residualization that settles it is Sec. IV-D. With these, 11 of 12 learned axes have cleared the gate on *transfer*; Sec. IV-D separates transfer from specificity.

D. Bifactor Structure: One Strong Factor Plus Separable Specifics

Clearing the cross-dataset gate establishes that an axis carries real, generalizing signal; it does *not* establish that the signal is the axis’s *own*, rather than a general good/bad valence shared across axes. A pre-registered bifactor readout—registered before the general-valence encoder G was trained and before any loading was fit—separates the two, and its verdict is decisive.

G is real (Phase A1). A family-pooled signed-valence encoder clears the same baseline-relative gate as every named axis: cross-dataset AUROC 0.856 ± 0.008 ($n = 51,319$, three seeds), +0.337 over its maximum null (untrained 0.480, bag-of-words 0.519). The general good/bad factor is not an artifact of any single corpus.

Most named axes are G in a coat (Phase A2). Residualizing each specific axis against G and re-measuring cross-corpus transfer, the pre-registration required ≥ 8 of 11 axes to remain diagonal-dominant. *Four did* (Table II). Five axes—purity, legitimacy, loyalty, care, fairness—are ≥ 0.98 predictable from G on independent corpora: their real, gate-passing AUROCs are substantially *general valence*, not their named dimension. The genuinely specific axes are privacy, autonomy, and environmental, with physical harm’s residual surviving. The transfer matrix’s off-diagonal mean falls only $0.726 \rightarrow 0.527$ under residualization while its diagonal falls $0.954 \rightarrow 0.779$: the shared factor was carrying most of the apparent cross-axis structure.

The standing gate confirms it independently. Running the specificity gate—each axis scored against G on its own held-out pairs, registered margin 0.05—reproduces the split through independent computation: own-axis environmental +.31, privacy +.28, *identity_attack* +.25, autonomy +.17, physical +.08; demote-to- G purity −.12, epistemic −.07, legitimacy −.05, care −.04, loyalty −.02, fairness −.02. The G row is bimodal—0.85–0.94 on the demoted axes’ own pairs, chance (0.43–0.51) on the specific ones—the residualization result restated as a gate. The two methods diverge on exactly one axis, recorded not adjudicated: *identity_attack* is decisively specific by the gate (+.25) but mixed by A2 residualization (G -share 0.67).

What this binds. The pre-registration’s demotion rule is adopted: a channel that fails specificity is displayed as G , never as a hollow named axis. Two consequences for this

TABLE II
BIFACTOR READOUT (PHASE A2): HOW MUCH OF EACH NAMED AXIS IS GENERAL VALENCE G (G_{PRED} , G ’S CROSS-CORPUS PREDICTION OF THE AXIS), AND WHETHER THE RESIDUAL STILL BEST PREDICTS ITS OWN AXIS. PRE-REGISTRATION REQUIRED ≥ 8 OF 11 DIAGONAL-DOMINANT; FOUR ARE.

Axis	G -share	After residual
purity	0.9997	is G
legitimacy	0.9986	is G
loyalty	0.9900	is G
care	0.9873	is G
fairness	0.9836	is G
epistemic	0.9396	mostly G
physical harm	0.8197	residual survives
<i>identity_attack</i>	0.6658	mixed
privacy	0.6421	specific
environmental	0.5538	specific
autonomy	0.4945	specific

section’s claims. First, the “11 of 12” count of Sec. IV-C measures validated *transfer*; the count with validated *specificity* is smaller—five own-axis plus the surviving physical residual. Second, the coordinate system is, on these corpora, one strong valence factor plus a small number of separable specifics—consistent with the effective rank ≈ 5.7 of 9 measured in Sec. IV-E, the very rank collapse that motivated the pre-registration. This is a finding about these corpora as sampled, labeled as such, not a claim that moral space is one-dimensional: the specific axes are genuinely separable, and the instrument now reports *which*.

E. Which Axes Are Robust Measurement Targets?

A validation regime should report not only which axes pass but *what kind of signal* each axis is. A λ -sweep of domain-adversarial training (gradient-reversal strength $\lambda = 0$ vs. $\lambda = 1$) across the encoder suite produced a clean dissociation:

TABLE III
VALENCE IS REGISTER-INVARIANT; FOUNDATION *presence* IS REGISTER-BOUND. CROSS-DATASET STRUCTURE AUROC AT ADVERSARY OFF/ON.

Encoder	$\lambda = 0$	$\lambda = 1$	Verdict
<i>Valence (signed judgment)</i>			
General moral valence G	0.856	0.850	invariant (3/3 seeds)
Loyalty valence	0.911	0.892	invariant
Purity valence	0.811	0.801	invariant
<i>Foundation presence (is the concern at stake?)</i>			
Purity presence	0.719	0.699	passes; adversary hurts
Loyalty presence	0.661	0.641	loses pass under adversary
Care presence	0.574	0.585	fails both
Legitimacy presence	0.585	0.544	fails both
Fairness presence	0.569	0.499	stripped to chance

The adversary works (domain accuracy driven to 0.12, near chance) yet degrades all five presence encoders while leaving valence essentially untouched. Two further methods agree: a frozen-encoder angular-mode-removal probe (removing dominant shared principal components and renormalizing to the sphere) lifts no presence axis by ≥ 0.05 ; and the register/domain direction *is* the top principal component of

the presence representations ($|PC1 \cdot d_{\text{domain}}| = 0.97\text{--}1.00$)—the identity signal for care, fairness, and legitimacy is *entangled within* the register subspace, not orthogonal to it. The engineering decision follows the measurement: ship purity (robust) and loyalty ($\lambda = 0$ only) presence channels via plain joint-contrastive training, no adversary; and report the care/fairness/legitimacy presence axes as register-bound, i.e. not yet trustworthy cross-register instruments.

For the geometry itself, an effective-rank analysis of the feeder suite gives 5.68 of 9 (pre-registered orthogonal prediction 6.08; maximum pairwise correlation 0.34): the axes are partially independent, and the deviation from the prediction is a property of the *data*—causally removable—not a claim about moral space.

V. THE ATTACK: SCALAR AND VERDICT-LEVEL

We summarize the attack results of [1] that this article defends against; full protocols and per-model tables are in the conference paper.

Selective displacement. Linguistic framing, emotional anchoring, and irrelevant sensory detail displace judgments in every model tested (emotional anchoring $d = 0.60\text{--}1.06$; euphemistic rewriting -10 to -16 points on the 0–70 scale); gender swap and evaluation order do not. The common thread of the live surfaces is *salience manipulation*.

Threshold evasion. Against a moderation filter flagging total harm $> T$, euphemistic rewriting (morally invariant by construction) lowers panel-averaged harm by 14.0 points and silently reclassifies three of six audited gold items FLAG→PASS. The asymmetry—evasion displacement is twice exaggeration displacement—is the dangerous direction.

The exploit survives a principled kernel. Routing scenarios through the DEME v3 rule kernel (LLM-extracted ethical facts → GenevaEMV3 → typed verdict), euphemistic rewriting flips FORBID/AVOID to permissive in 13.7% of 161 cases; FORBID verdicts fall 44 → 27. A rule engine inherits, rather than cures, its front-end’s salience vulnerability. Hardening must target perception—which is what the remainder of this article does.

[TODO: Per-track verdict-level attack table from the native 9-D/10-D re-run (Sec. IX): displacement per DEME axis and verdict-flip rate per perturbation type, replacing the single aggregate 13.7% figure.]

VI. THE DEFENSE: EQUIVALENCE-CLASS AVERAGING

The attack exploits the fact that a single description is one arbitrary representative of an equivalence class of morally identical texts. The defense makes the decision a function of the *class*, not the representative: generate the input’s paraphrase class (m paraphrases), average per-dimension perception scores over the class, then decide. This is semantic randomized smoothing [12], [13] with a meaning-preserving perturbation family.

A. Mechanism Selection and the Two-Number Rule

We define decision-layer drift θ_d as the mean per-dimension decision movement under re-description, and pre-registered the

bar $\theta_d \leq 0.5$. Two candidate mechanisms were compared on the demonstration scenarios under a leave-scenario-out proxy: equivalence-class averaging reached $\theta_d = 0.42$ (reframe) / 0.39 (euphemism), while drift-subspace projection—removing the top principal component of embedding paraphrase-differences—failed even in-sample (0.52): the score-moving drift is not the top PC of paraphrase variation. Averaging was *selected*; we emphasize (two-number rule) that 0.42 is mechanism-selection evidence only, because the bar was calibrated from the same smoke scenarios. Bar-meeting is established by the at-scale run:

B. At-Scale Result

Pre-registered split-half protocol: 60 held-out civil-comments items (disjoint by SHA-256 of normalized text from all encoder training rows), paraphrase classes of $m = 6$ generated by an open instruction model, scored through the full 9-feeder perception layer. Result: raw drift 0.407 → averaged $\theta_d = 0.219$ —the mechanism halves salience drift and meets the pre-registered bar.

Two caveats are part of the record. First, the at-scale paraphrases are *natural*, not adversarial (raw drift 0.407 vs. 0.67–0.85 for the hand-built adversarial demo transforms); adversarial robustness at scale is a distinct, still-open measurement. **[TODO: adversarial-paraphrase θ_d at scale (containment item).]** Second, the LLM paraphrase generator *refused* 24% of harmful inputs (19/79 toxic items), producing singleton classes on exactly the content that matters.

C. Red-Team: the Hole Belongs to the Generator

To test whether the refusal hole is intrinsic to the mechanism, we replaced the LLM paraphraser with a non-refusing red-team instrument: NLLB back-translation [22] through six pivot languages (es/fr/de/zh/ar/ru), which refused 0% of the same items (the LLM refused 20% of them). On 50 fresh high-toxicity items (toxicity ≥ 0.7): raw drift 0.482 → $\theta_d = 0.301 \leq 0.5$. The 24% hole is the *generator’s*, not the *mechanism’s*: a non-refusing paraphraser closes it. Back-translation drifts meaning slightly, so we class it as a red-team instrument rather than necessarily the production paraphraser.

D. Trust Boundary and Auditability

Generating the class at inference has three consequences we make explicit. *Cost*: perception cost multiplies by m ; throughput must be measured with averaging on. *Security*: the paraphrase generator joins the trust boundary—an attacker who can starve it (provoke refusal) reduces the class to a singleton and recovers the raw, weakly-invariant score. The deployed policy is therefore *refusal* → *flagged singleton* → *escalate by default*. *Auditability*: the decision’s hash-chained audit proof records the class member hashes, class size, and refusal flag, so the invariance evidence behind every decision is re-verifiable.

VII. FROM ESCALATION TO MODERATION: THE LEARNED CONTRACTION

An instrument that only escalates is safe but useless. The pipeline moderates—emits allow/remove—exactly where a learned contraction over the validated channels is itself validated, and escalates elsewhere.

A logistic contraction over the nine validated feeders (the eight original passing axes plus `identity_attack`) reaches out-of-fold AUROC **0.863** / F1 0.76, a +0.084 lift over the 8-feeder baseline (0.779); `identity_attack` carries the largest weight (−2.69). Decision thresholds ($p_{\text{remove}} = 0.50$, $p_{\text{allow}} = 0.10$, validation bar 0.70) are frozen into a versioned artifact consumed by the decision layer: on the balanced evaluation set the pipeline moderates ~51% of items with ~80% out-of-fold remove-precision and escalates the rest.

Leakage control. The `identity_attack` encoder was trained partly on civil-comments rows, and it is the contraction’s dominant feature; if fit rows overlapped encoder-training rows, the lift would be self-scoring. We measured the overlap (77/1600 rows, 4.8%, by SHA-256 of normalized text) and refit on the 1,523 disjoint rows: AUROC 0.872 → 0.863, lift +0.089 → +0.084, weight −2.78 → −2.69. The lift is real signal. The fit harness now enforces disjointness. One honest false positive (harsh criticism → remove) is kept in the record; we declined to tune the threshold to the demonstration scenarios.

[TODO: Like-for-like baseline: decision-vs-decision comparison against a scalar-toxicity moderation baseline (Detoxify/Perspective-class) under identical inputs and identical euphemistic attack, reporting flip rates for both. The scalar baseline’s drift is measured; the decision-level comparison table is not yet run.]

VIII. CROSS-LINGUAL INVARIANCE

[1] claimed English-only scope. The extended instrument locates invariance architecturally and measures it across languages.

Invariance lives in the canonicalization layer. Raw feeder scores are only weakly invariant under re-description (invariance ratio 0.60): trained axes *amplify* embedding drift. Canonicalizing to an anchor representation before scoring yields same-language invariance index 0.75 [0.64, 0.84]. Encoder choice matters and was measured, not assumed: BGE-M3 is the production canonicalizer (LaBSE wins cross-lingually but collapses same-language structure to 0.49).

At scale, across scripts, harmful ≈ benign. On 60 items (half harmful) translated to es/ar/zh/hi/sw via NLLB [22]: invariance index 0.721 [0.71, 0.74] (BGE-M3) / 0.804 [0.79, 0.82] (LaBSE). Critically, harmful and benign items are indistinguishable (BGE 0.704/0.739; LaBSE 0.797/0.811): the invariance is not a benign-only artifact, and the small- n demonstration values (0.74/0.81) are confirmed with tight intervals. An operational finding with security content: chat LLMs *refuse to translate* harmful content (the same failure mode as paraphrase refusal, Sec. VI-C); the translation legs therefore run on NLLB, a non-refusing dedicated MT model.

IX. NATIVE RE-RUN OF THE VULNERABILITY TRACKS

[TODO: This section is the main outstanding experiment (planned August 2026, NRP open panel). Design, pre-registered here: (1) each (scenario × model × perturbation) case from the five conference tracks is scored natively in the DEME 9-D (+identity_attack) space through the erisml-compiler pipeline, yielding a MoralTensorV3 and a DEMEVERdict per case; (2) displacement is measured per axis and at the verdict level per track, superseding the scalar-total results; (3) the compiler’s framework-pluralist cross-projection disagreement surface is tested as an attack detector: does disagreement light up under salience manipulation? (4) the 7-D→9-D mapping table of [1] is retained for comparability. Models judge in native 9-D rather than post-hoc mapping.]

X. DISCUSSION

The arc closes. The conference paper’s finding was uncomfortable: salience manipulation displaces every model tested, evasion is easier than tripping, and principled downstream kernels inherit the vulnerability. The extended results give the deployment answer: measure in a validated, falsifiable geometry (Sec. IV); decide on equivalence classes rather than representatives, with the residual attack surface (generator starvation) explicitly escalated and audited (Sec. VI); moderate only where the contraction is validated (Sec. VII); and carry the whole case in a re-verifiable audit proof.

What kind of claim the geometry makes. The discovery scorecard (validated / retracted / declined), the bifactor separation of shared valence from axis-specific signal, and the valence–presence dissociation change the epistemic status of the coordinate system. The dimension set is not a taxonomy we assert; it is an empirical object with an admission gate, a falsification record, a measured factor structure, and a characterization of which axes are cross-register trustworthy. The bifactor result is the sharpest of these: most nominal axes are, on these corpora, one strong valence factor in different coats, and the instrument now reports specificity rather than merely transfer. When an axis is not trustworthy (rights; care/fairness/legitimacy presence) or not specific (the demoted five), the instrument says so—displaying G , or escalating—rather than scoring a hollow coordinate.

Limits. (i) The defense is measured against natural paraphrase drift and one non-refusing red-team generator; adversarial-paraphrase robustness at scale is open (Sec. VI-B). (ii) The learned-encoder results validate the perception layer, not the five conference tracks re-run natively; that experiment is designed but not run (Sec. IX). (iii) Corpus norms: moderation corpora are predominantly English and North-American; the cross-lingual result covers re-description of the same content, not cross-cultural moral variation. (iv) The purity second-corpus leg is thin (Sec. IV-C); its AUROC is provisional. (v) Sample sizes for the worked exploit remain an existence proof ($n = 6$ gold), with displacement magnitudes established on larger sets. (vi) The bifactor separation (Sec. IV-D) is measured on the present moderation and social-norm corpora; whether the strong shared valence factor reflects moral space

or these corpora’s sampling of it is not settled here, and the demotion it drives is a property of the instrument on this data, not a claim that the demoted concerns lack independent moral content.

XI. CONCLUSION

We extended a measured attack on LLM moral judgment into a validated, defended, auditable moderation instrument. The moral geometry is no longer merely reliable but extensible and falsifiable—one dimension discovered and validated cross-dataset, one retracted, one declined—and its axes are characterized as register-invariant (valence) or register-bound (presence) by three agreeing methods. Equivalence-class averaging halves decision-layer drift at scale under a pre-registered bar, with the residual hole located in the paraphrase generator and closed by a non-refusing red-team instrument. A leakage-controlled contraction converts escalation into moderation where validated. Invariance is confirmed cross-lingually at scale with harmful $\approx$ benign. What remains—the native re-run of the five vulnerability tracks through this instrument—is designed, pre-registered, and scheduled; it will close the loop between auditing deployed judges and deploying a judge built to pass the audit.

REPRODUCIBILITY

All post-conference experiments are public: the Moral Spectrum Analyzer pipeline (`pip install moral-spectrum-analyzer`) [4], the validated per-dimension encoders and gate (`pip install xbse`) [3], and the DEME v3 kernel (`erisml-compiler` [2]). Pre-registrations, prediction files, discovery records (including the retraction and the declination), λ -sweep artifacts, invariance and cross-lingual data, and the contraction with its leakage-control hashes are in the respective repositories. Conference-track code and data remain in the `agi-hpc` package (tag `bds2026-v2`), archived on Zenodo [1].

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REFERENCES

- [1] A. H. Bond, “Selective invariance violations in large language model moral judgment: A geometric framework for behavioral manipulation detection,” in *Proc. IEEE Int. Conf. Big Data Computing Service and Machine Learning Applications (BigDataService)*, 2026. Artifact: Zenodo, doi:10.5281/zenodo.20660065.
- [2] A. H. Bond, *ErisML Compiler: A Framework-Pluralist Compiler from Natural-Language Moral Material to a Typed Moral Graph*, software, v0.9, 2026. Zenodo, doi:10.5281/zenodo.20659432.
- [3] A. H. Bond, *xbse: Modular Domain-Specific Sentence Embeddings, Validated by One Shared Gate*, software, v0.1.0, 2026. [Online]. Available: <https://github.com/ahb-sjsu/xbse>
- [4] A. H. Bond, *Moral Spectrum Analyzer*, software, v0.1.0, 2026. [Online]. Available: <https://github.com/ahb-sjsu/moral-spectrum-analyzer>
- [5] M. Sharma *et al.*, “Towards understanding sycophancy in language models,” in *Proc. ICLR*, 2024.
- [6] E. Perez *et al.*, “Discovering language model behaviors with model-written evaluations,” in *Findings of ACL*, 2023.
- [7] A. Tversky and D. Kahneman, “The framing of decisions and the psychology of choice,” *Science*, vol. 211, pp. 453–458, 1981.
- [8] J. M. Echterhoff, Y. Liu, and A. Alessa, “Cognitive biases in large language models: A survey and mitigation framework,” *arXiv:2410.02466*, 2024.
- [9] N. Scherrer *et al.*, “Evaluating the moral beliefs encoded in LLMs,” in *Proc. NeurIPS*, 2023.
- [10] A. H. Bond, *Geometric Methods in Computational Modeling: From Manifolds to Production Systems*. San Jose State University, 2026.
- [11] L. Thiele, “Does LaBSE encode moral geometry? Testing Bond’s geometric ethics framework through cross-lingual probing,” Undergraduate Research Report, UCLA Dept. of Cognitive Science, 2026.
- [12] J. Cohen, E. Rosenfeld, and Z. Kolter, “Certified adversarial robustness via randomized smoothing,” in *Proc. ICML*, pp. 1310–1320, 2019.
- [13] A. Robey, E. Wong, H. Hassani, and G. J. Pappas, “Smooth-LLM: Defending large language models against jailbreaking attacks,” *arXiv:2310.03684*, 2023.
- [14] D. Borkan, L. Dixon, J. Sorensen, N. Thain, and L. Vasserman, “Nuanced metrics for measuring unintended bias with real data for text classification,” in *Companion Proc. WWW*, pp. 491–500, 2019.
- [15] C. J. Kennedy, G. Bacon, A. Sahn, and C. von Vacano, “Constructing interval variables via faceted Rasch measurement and multitask deep learning: a hate speech application,” *arXiv:2009.10277*, 2020.
- [16] J. Graham *et al.*, “Moral foundations theory: The pragmatic validity of moral pluralism,” *Adv. Exp. Soc. Psychol.*, vol. 47, pp. 55–130, 2013.
- [17] M. Forbes, J. D. Hwang, V. Schwartz, M. Sap, and Y. Choi, “Social chemistry 101: Learning to reason about social and moral norms,” in *Proc. EMNLP*, pp. 653–670, 2020.
- [18] D. Hendrycks *et al.*, “Aligning AI with shared human values,” in *Proc. ICLR*, 2021.
- [19] J. Trager *et al.*, “The Moral Foundations Reddit Corpus,” *arXiv:2208.05545*, 2022.
- [20] F. Feng *et al.*, “Language-agnostic BERT sentence embedding,” in *Proc. ACL*, pp. 878–891, 2022.
- [21] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” *arXiv:2402.03216*, 2024.
- [22] NLLB Team *et al.*, “No language left behind: Scaling human-centered machine translation,” *arXiv:2207.04672*, 2022.